

Reinforcement Learning for Single-Qubit Quantum State Preparation

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AI Models for Physics

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1. Introduction

This project is an implementation of the article *Reinforcement Learning in Different Phases of Quantum Control* of Bukov et al. [1]. The goal is to prepare a physical system in a desired quantum state. It is a task central to many areas of physics. Preparing them quickly and with high fidelity is still an hard challenge. For long protocol durations adiabatic evolution can be used to robustly reach the target quantum state, but it relies on assumptions that are often violated in real life applications.

The goal is to develop a Reinforcement Learning (RL) agent that learns the underlying physical system with only a scalar reward provided only at the last step of the episode, in the form of the fidelity between the current state and the target. The control field is restricted to bang-bang protocols, $h \in \{-h_{\max}, +h_{\max}\}$, for a finite time horizon, based on the terminal time T three phases are studied: overconstrained, glassy, and controllable.

This project includes an implementation of the Bukov et al. algorithm in the single qubit setting under bang-bang control. A simple Q-learning baseline as well as two DQN comparison to assess how the choice of learning algorithm affects performance across the three phases.

2. The physical model

The qubit follows the spin-1/2 Hamiltonian

$$H[h_x(t)] = -S^z - h_x(t) S^x$$

where S^α are the spin-1/2 operators. The control field is discrete $h_x(t) \in \{-4, +4\}$. The initial and target states are the ground states of H at $h_x = -2$ and $h_x = +2$. The goal is to maximize the final fidelity after a protocol of duration T :

$$F_h(T) = |\langle \psi_* | \psi(T) \rangle|^2$$

under unitary evolution. Finding the best protocol is a search in the space made of all the possible 2^{N_T} protocols.

3. RL setup

The MDP is defined by the tuple (S, A, p, R, N_T) . The state is $s = (t, h_x(t)/h_{\max})$, the action is the field choice $a \in \{-h_{\max}, +h_{\max}\}$, and the transition is the unitary evolution $|\psi'\rangle = e^{-iH[h]dt}|\psi\rangle$. The reward is sparse: zero at every step except the last, where $r = F_h(T)$. Each episode has a fixed length N_T , with time step $dt = T/N_T$. No agent is provided with any physical knowledge of the system.

4. Methods

Four different methods have been developed. First a simple stochastic descent algorithm that explores the search space with a local search to compute the optimum. Since the action space is relatively simple this is still computationally feasible.

Then a implementation of the Watkins $Q(\lambda)$ with replay, used in the Bukov et al. paper. Followed by a simple Q -learning baseline.

And finally, two DQN approaches are compared: one using a basic DQN and one modified with the same features as the Bukov et al. paper.

The most important addition made by Bukov et al. is the eligibility trace, since the reward is sparse for standard Q -learning it is hard to propagate the signal back to the early pulses. Traces allow it to be propagated back in a single sweep, with a decay parameter λ controlling how far credit reaches across the episode. Traces are only updated when following the greedy action to avoid violating the off-policy assumption of Watkins $Q(\lambda)$, otherwise they are cut to zero.

Another important addition of the paper is the forced learning. The algorithm replays the best protocol found so far and stores it in the Q -table, alternating with the standard learning steps. This ensures the agent never drifts too far from the best policy found so far and actively tries to improve it.

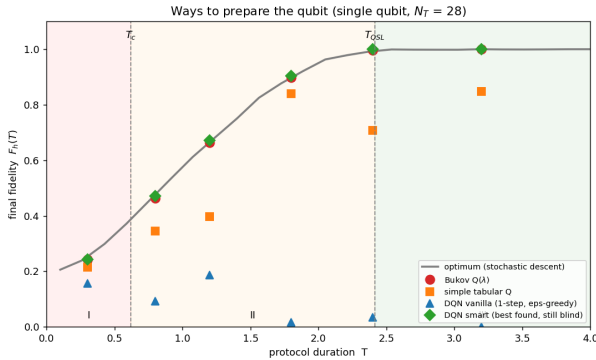
The last element is softmax exploration with a β ramp that grows during training, gradually shifting from random to greedy, mimicking ϵ -greedy but with a smoother transition. The temporal-difference update is

$$Q(s_i, a_i) \leftarrow Q(s_i, a_i) + \alpha \left[r_i + \max_a Q(s_{i+1}, a) - Q(s_i, a_i) \right].$$

5. Results

The phase diagram shows the best achievable bang-bang fidelity $F_h(T)$ as a function of protocol duration T , recovered via stochastic descent. Three regimes are visible: for $T < T_c \approx 0.6$ the problem is overconstrained and fidelity is limited regardless of the protocol. For $T_c < T < T_{QSL} \approx 2.4$ the landscape is glassy and finding the optimum is hard. Past the quantum speed limit T_{QSL} unit fidelity becomes reachable. The quantum speed limit is the minimum time required to drive the system to the target state given the field constraint $|h_x| \leq h_{\max}$, below it no protocol can achieve $F = 1$. The stochastic descent recovers $F = 0.576$ at $T = 1.0$, matching the value reported in the paper.

The comparison table shows all methods sharing the same physics-blind (t, h_x) observation, so differences are purely due to the learning algorithm. The Bukov $Q(\lambda)$ agent tracks the optimum closely across all phases. Simple Q -learning degrades in the glassy phase, where the one-step update cannot propagate the sparse reward effectively. DQN base collapses entirely, confirming that vanilla deep RL cannot handle this observation space without the mechanisms introduced by Bukov et al. The DQN smart variant, which includes the replay and exploration schedule, reach performances close to the optimum. At $T = 2.4$ both Bukov Q and DQN smart surpass the stochastic optimum. This is reasonable since its not an exhaustive search with optimality guarantees.



T	stoc-opt	simple Q	Bukov Q	DQN base	DQN smart
0.3	0.251	0.215	0.242	0.156	0.243
0.8	0.477	0.346	0.462	0.093	0.472
1.2	0.669	0.396	0.664	0.187	0.671
1.8	0.899	0.839	0.897	0.017	0.904
2.4	0.993	0.707	0.997	0.034	0.999
3.2	1.000	0.849	1.000	0.000	1.000

Figure 2: Final fidelity by method and protocol duration.

Figure 1: The four methods compared at fixed $N_T = 28$

7. Conclusions

The goal of Bukov et al. was to build an RL agent that, without any physical knowledge of the system, is able to learn optimal control protocols and aid in the construction of variational theories. In the single-qubit case, inspecting the best protocol found by the agent reveals a simple structure that can be summarized by a single variational parameter. In this project the addition of a plain DQN and a modified DQN with the same mechanisms as Bukov et al. makes the role of each algorithmic choice explicit: without eligibility traces, forced learning, and the softmax schedule, even a neural network agent fails completely. DQN smart and Bukov $Q(\lambda)$ perform comparably across the experiments conducted, suggesting the key ingredients are the training mechanisms rather than the function approximator. Future work could explore the multi-qubit setting and how these approaches scale with system size.

References

- [1] Marin Bukov et al. “Reinforcement Learning in Different Phases of Quantum Control”. In: *Physical Review X* 8.3 (2018). ISSN: 2160-3308. DOI: 10.1103/physrevx.8.031086. URL: <http://dx.doi.org/10.1103/PhysRevX.8.031086>.